CS2 Dataset Overview and Initial Analysis

Sections 2, 2.1, 2.2, 2.3, 3 and 4

Setup

This report reads the three Parquet datasets, prints descriptive tables directly into the rendered report, and saves figures as PDF files in `../reports/figures`.

```
library(tidyverse)
library(arrow)
library(knitr)

# Paths are relative to the directory containing this QMD file.
rounds_file <- "../data/analysis/rounds.parquet"
states_file <- "../data/analysis/round_states.parquet"
enriched_file <- "../data/analysis/round_states_enriched.parquet"

figures_dir <- "../reports/figures"
dir.create(figures_dir, showWarnings = FALSE, recursive = TRUE)

rounds <- read_parquet(rounds_file) |> as_tibble()
round_states <- read_parquet(states_file) |> as_tibble()
round_states_enriched <- read_parquet(enriched_file) |> as_tibble()

stopifnot(all(c("match_id", "map", "round_num") %in% names(rounds)))
stopifnot(all(c("match_id", "map", "round_num", "tick", "seconds",
                "ct_alive", "t_alive", "ct_win") %in% names(round_states)))
stopifnot(all(c("match_id", "map", "round_num") %in%
                names(round_states_enriched)))

# Save each ggplot as a PDF.
save_pdf <- function(plot, filename, width = 9, height = 6) {
  ggsave(
    filename = file.path(figures_dir, filename),
```

```
    plot = plot,
    width = width,
    height = height,
    device = grDevices::cairo_pdf
  )
}

# Consistent table display in the rendered report.
show_table <- function(x, digits = 2) {
  knitr::kable(x, digits = digits, format = "html")
}
```

2. Dataset structure

2.1 Overview of the three datasets

```
dataset_overview <- tibble(
  dataset = c("rounds", "round_states", "round_states_enriched"),
  rows = c(nrow(rounds), nrow(round_states), nrow(round_states_enriched)),
  columns = c(ncol(rounds), ncol(round_states), ncol(round_states_enriched)),
  missing_cells = c(sum(is.na(rounds)), sum(is.na(round_states)),
                    sum(is.na(round_states_enriched))),
  unique_matches = c(n_distinct(rounds$match_id),
                     n_distinct(round_states$match_id),
                     n_distinct(round_states_enriched$match_id)),
  unique_rounds = c(
    nrow(distinct(rounds, match_id, round_num)),
    nrow(distinct(round_states, match_id, round_num)),
    nrow(distinct(round_states_enriched, match_id, round_num))
  )
)
show_table(dataset_overview)
```

dataset	rows	columns	missing_cells	unique_matches	unique_rounds
rounds	1257	26	1288	60	1257
round_states	116201	45	195710	60	1257
round_states_enriched	116201	59	250917	60	1257

2.2 Column names and data types

```
column_overview <- bind_rows(
  tibble(dataset = "rounds", column = names(rounds),
         data_type = map_chr(rounds, ~ paste(class(.x), collapse = "/"))),
  tibble(dataset = "round_states", column = names(round_states),
         data_type = map_chr(round_states, ~ paste(class(.x), collapse = "/"))),
  tibble(dataset = "round_states_enriched", column = names(round_states_enriched),
         data_type = map_chr(round_states_enriched,
                             ~ paste(class(.x), collapse = "/")))
)
show_table(column_overview)
```

dataset	column	data_type
rounds	match_id	character
rounds	map	character
rounds	part	integer
rounds	round_num	integer
rounds	is_overtime	logical
rounds	is_pistol_round	logical
rounds	ct_team	character
rounds	t_team	character
rounds	ct_score_before	integer
rounds	t_score_before	integer
rounds	ct_won_previous_round	logical
rounds	ct_buy_type	character
rounds	t_buy_type	character
rounds	ct_equipment_freeze_end	integer
rounds	t_equipment_freeze_end	integer
rounds	bomb_planted	logical
rounds	plant_site	character
rounds	first_death_side	character
rounds	freeze_end_tick	integer
rounds	plant_tick	integer
rounds	end_tick	integer
rounds	round_duration_s	numeric
rounds	winner_side	character
rounds	winner_team	character
rounds	ct_win	logical
rounds	reason_name	character

dataset	column	data_type
round_states	match_id	character
round_states	map	character
round_states	part	integer
round_states	round_num	integer
round_states	tick	integer
round_states	seconds	numeric
round_states	time_remaining_s	numeric
round_states	bomb_planted	logical
round_states	seconds_since_plant	numeric
round_states	plant_site	character
round_states	bomb_carried	logical
round_states	ct_team	character
round_states	t_team	character
round_states	ct_score_before	integer
round_states	t_score_before	integer
round_states	ct_alive	integer
round_states	ct_hp	integer
round_states	ct_armor	integer
round_states	ct_helmets	integer
round_states	ct_defusers	integer
round_states	ct_equipment	integer
round_states	ct_money	integer
round_states	ct_rifles	integer
round_states	ct_awps	integer
round_states	ct_smgs	integer
round_states	ct_smokes	integer
round_states	ct_flashes	integer
round_states	ct_he	integer
round_states	ct_molotovs	integer
round_states	t_alive	integer
round_states	t_hp	integer
round_states	t_armor	integer
round_states	t_helmets	integer
round_states	t_defusers	integer
round_states	t_equipment	integer
round_states	t_money	integer
round_states	t_rifles	integer
round_states	t_awps	integer
round_states	t_smgs	integer
round_states	t_smokes	integer
round_states	t_flashes	integer

dataset	column	data_type
round_states	t_he	integer
round_states	t_molotovs	integer
round_states	winner_side	character
round_states	ct_win	logical
round_states_enriched	match_id	character
round_states_enriched	map	character
round_states_enriched	part	numeric
round_states_enriched	round_num	numeric
round_states_enriched	tick	numeric
round_states_enriched	seconds	numeric
round_states_enriched	time_remaining_s	numeric
round_states_enriched	bomb_planted	logical
round_states_enriched	seconds_since_plant	numeric
round_states_enriched	plant_site	character
round_states_enriched	bomb_carried	logical
round_states_enriched	ct_team	character
round_states_enriched	t_team	character
round_states_enriched	ct_score_before	numeric
round_states_enriched	t_score_before	numeric
round_states_enriched	ct_alive	numeric
round_states_enriched	ct_hp	numeric
round_states_enriched	ct_armor	numeric
round_states_enriched	ct_helmets	numeric
round_states_enriched	ct_defusers	numeric
round_states_enriched	ct_equipment	numeric
round_states_enriched	ct_money	numeric
round_states_enriched	ct_rifles	numeric
round_states_enriched	ct_awps	numeric
round_states_enriched	ct_smgs	numeric
round_states_enriched	ct_smokes	numeric
round_states_enriched	ct_flashes	numeric
round_states_enriched	ct_he	numeric
round_states_enriched	ct_molotovs	numeric
round_states_enriched	t_alive	numeric
round_states_enriched	t_hp	numeric
round_states_enriched	t_armor	numeric
round_states_enriched	t_helmets	numeric
round_states_enriched	t_defusers	numeric
round_states_enriched	t_equipment	numeric
round_states_enriched	t_money	numeric
round_states_enriched	t_rifles	numeric

dataset	column	data_type
round_states_enriched	t_awps	numeric
round_states_enriched	t_smgs	numeric
round_states_enriched	t_smokes	numeric
round_states_enriched	t_flashes	numeric
round_states_enriched	t_he	numeric
round_states_enriched	t_molotovs	numeric
round_states_enriched	winner_side	character
round_states_enriched	ct_win	logical
round_states_enriched	is_overtime	logical
round_states_enriched	is_pistol_round	logical
round_states_enriched	ct_won_previous_round	logical
round_states_enriched	ct_buy_type	character
round_states_enriched	t_buy_type	character
round_states_enriched	ct_equipment_freeze_end	numeric
round_states_enriched	t_equipment_freeze_end	numeric
round_states_enriched	first_death_side	character
round_states_enriched	freeze_end_tick	numeric
round_states_enriched	plant_tick	numeric
round_states_enriched	end_tick	numeric
round_states_enriched	round_duration_s	numeric
round_states_enriched	winner_team	character
round_states_enriched	reason_name	character

2.3 Maps and teams

```
map_overview <- rounds |>
  group_by(map) |>
  summarise(rounds = n(), matches = n_distinct(match_id), .groups = "drop")
show_table(map_overview)
```

map	rounds	matches
de_dust2	628	30
de_mirage	629	30

```
team_overview <- bind_rows(
  rounds |> transmute(team = ct_team),
  rounds |> transmute(team = t_team)
```

```
) |>
  filter(!is.na(team)) |>
  distinct(team) |>
  arrange(team)
show_table(team_overview)
```

team
9z
Aurora
FURIA
FUT
FaZe
Falcons
G2
Legacy
MOUZ
NAVI
Spirit
Vitality

2.1 Round-level data: `rounds.parquet`

Each row is expected to represent one round.

Duplicate round keys

```
round_key_check <- rounds |>
  count(match_id, round_num, name = "rows_per_round") |>
  filter(rows_per_round > 1)

tibble(duplicate_round_keys = nrow(round_key_check)) |> show_table()
```

duplicate_round_keys
0

```r
if (nrow(round_key_check) > 0) show_table(round_key_check)
```

Round summary by map

```r
rounds_by_map <- rounds |>
  group_by(map) |>
  summarise(
    rounds = n(),
    matches = n_distinct(match_id),
    pistol_rounds = sum(is_pistol_round %in% TRUE, na.rm = TRUE),
    overtime_rounds = sum(is_overtime %in% TRUE, na.rm = TRUE),
    planted_rounds = sum(bomb_planted %in% TRUE, na.rm = TRUE),
    planted_pct = 100 * mean(bomb_planted %in% TRUE, na.rm = TRUE),
    mean_duration_s = mean(round_duration_s, na.rm = TRUE),
    median_duration_s = median(round_duration_s, na.rm = TRUE),
    .groups = "drop"
  )
show_table(rounds_by_map)
```

map	rounds	matches	pistol_rounds	overtime_rounds	planted_rounds	planted_pct	mean_d
de_dust2	628	30	60	14	349	55.57	
de_mirage	629	30	60	16	312	49.60	

Match length

```r
rounds_per_match <- rounds |>
  distinct(match_id, map, round_num) |>
  count(match_id, map, name = "rounds_in_match")

match_length_summary <- rounds_per_match |>
  summarise(
    matches = n(),
    min_rounds = min(rounds_in_match),
    q1_rounds = quantile(rounds_in_match, 0.25),
    median_rounds = median(rounds_in_match),
    mean_rounds = mean(rounds_in_match),
    q3_rounds = quantile(rounds_in_match, 0.75),
```

```
    max_rounds = max(rounds_in_match)
  )
show_table(match_length_summary)
```

matches	min_rounds	q1_rounds	median_rounds	mean_rounds	q3_rounds	max_rounds
60	14	18.75	21	20.95	22.25	35

Round outcomes by map and winning side

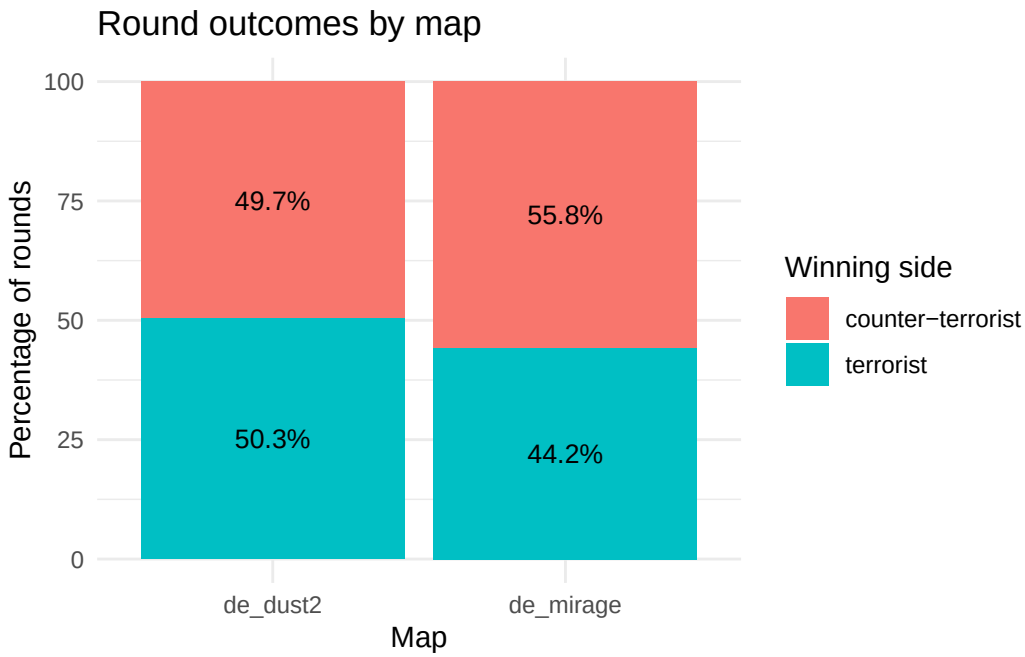
```
round_outcomes <- rounds |>
  count(map, winner_side, name = "rounds") |>
  group_by(map) |>
  mutate(percentage = 100 * rounds / sum(rounds)) |>
  ungroup()

show_table(round_outcomes)
```

map	winner_side	rounds	percentage
de_dust2	counter-terrorist	312	49.68
de_dust2	terrorist	316	50.32
de_mirage	counter-terrorist	351	55.80
de_mirage	terrorist	278	44.20

```
p_round_outcomes <- ggplot(
  round_outcomes, aes(x = map, y = percentage, fill = winner_side)
) +
  geom_col() +
  geom_text(aes(label = sprintf("%.1f%%", percentage)),
            position = position_stack(vjust = 0.5), size = 3.5) +
  labs(title = "Round outcomes by map", x = "Map",
       y = "Percentage of rounds", fill = "Winning side") +
  theme_minimal()

p_round_outcomes
```




```
save_pdf(p_round_outcomes, "21_round_outcomes_by_map.pdf")
```

2.2 Round-state data: `round_states.parquet`

Each row is a snapshot of a round at a particular time. Snapshots from the same round are repeated observations, not independent rounds.

Snapshots per round

```
snapshots_per_round <- round_states |>
  count(match_id, map, round_num, name = "snapshots")

snapshot_count_summary <- snapshots_per_round |>
  summarise(
    rounds = n(),
    min_snapshots = min(snapshots),
    q1_snapshots = quantile(snapshots, 0.25),
    median_snapshots = median(snapshots),
    mean_snapshots = mean(snapshots),
    q3_snapshots = quantile(snapshots, 0.75),
```

```
    max_snapshots = max(snapshots)
  )
show_table(snapshot_count_summary)
```

rounds	min_snapshots	q1_snapshots	median_snapshots	mean_snapshots	q3_snapshots	max_snaps
1257	14	71	95	92.44	115	

Snapshot timing and missing values

```
snapshot_timing <- round_states |>
  summarise(
    min_seconds = min(seconds, na.rm = TRUE),
    median_seconds = median(seconds, na.rm = TRUE),
    max_seconds = max(seconds, na.rm = TRUE),
    min_time_remaining = min(time_remaining_s, na.rm = TRUE),
    max_time_remaining = max(time_remaining_s, na.rm = TRUE)
  )
show_table(snapshot_timing)
```

min_seconds	median_seconds	max_seconds	min_time_remaining	max_time_remaining
0	46	155	-0.98	115.2

```
snapshot_missingness <- tibble(
  column = names(round_states),
  missing_values = map_int(round_states, ~ sum(is.na(.x))),
  missing_pct = 100 * missing_values / nrow(round_states)
) |>
  arrange(desc(missing_values))
show_table(snapshot_missingness)
```

column	missing_values	missing_pct
seconds_since_plant	97855	84.21
plant_site	97855	84.21
match_id	0	0.00
map	0	0.00

column	missing_values	missing_pct
part	0	0.00
round_num	0	0.00
tick	0	0.00
seconds	0	0.00
time_remaining_s	0	0.00
bomb_planted	0	0.00
bomb_carried	0	0.00
ct_team	0	0.00
t_team	0	0.00
ct_score_before	0	0.00
t_score_before	0	0.00
ct_alive	0	0.00
ct_hp	0	0.00
ct_armor	0	0.00
ct_helmets	0	0.00
ct_defusers	0	0.00
ct_equipment	0	0.00
ct_money	0	0.00
ct_rifles	0	0.00
ct_awps	0	0.00
ct_smgs	0	0.00
ct_smokes	0	0.00
ct_flashes	0	0.00
ct_he	0	0.00
ct_molotovs	0	0.00
t_alive	0	0.00
t_hp	0	0.00
t_armor	0	0.00
t_helmets	0	0.00
t_defusers	0	0.00
t_equipment	0	0.00
t_money	0	0.00
t_rifles	0	0.00
t_awps	0	0.00
t_smgs	0	0.00
t_smokes	0	0.00
t_flashes	0	0.00
t_he	0	0.00
t_molotovs	0	0.00
winner_side	0	0.00

ct_win	0	0.00

Alive-player advantage and eventual outcome

This is a snapshot-level exploratory comparison. Because the same round contributes multiple snapshots, the percentages below should not be interpreted as independent-round estimates.

```
state_advantage <- round_states |>
  mutate(
    alive_difference = ct_alive - t_alive,
    alive_advantage = case_when(
      alive_difference > 0 ~ "CT advantage",
      alive_difference < 0 ~ "T advantage",
      alive_difference == 0 ~ "Equal alive players",
      TRUE ~ NA_character_
    )
  )

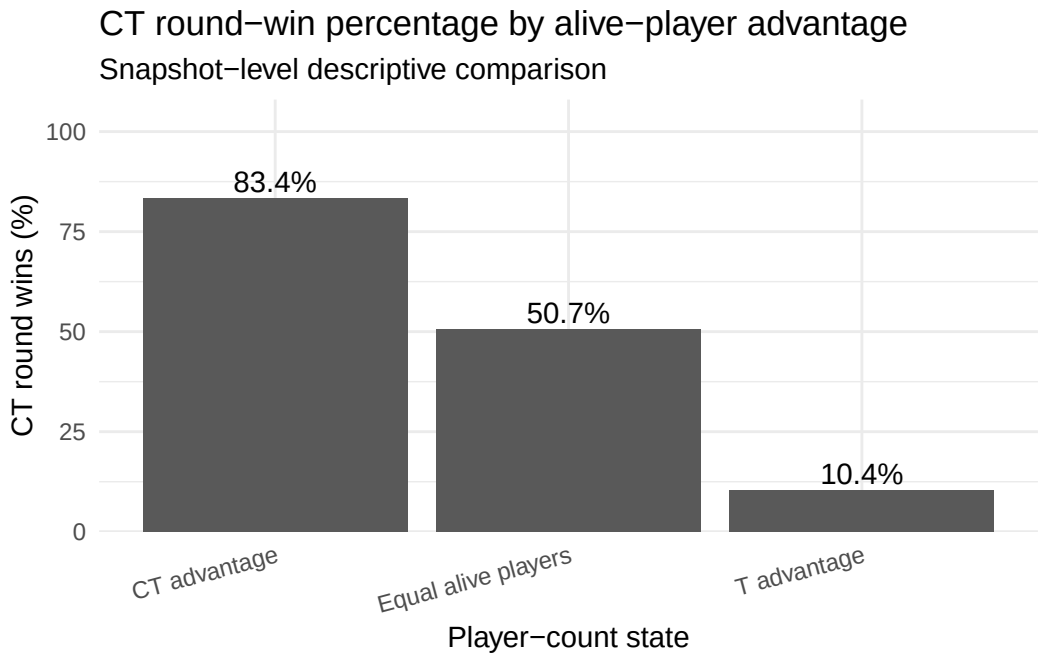
advantage_by_outcome <- state_advantage |>
  filter(!is.na(ct_win), !is.na(alive_advantage)) |>
  group_by(alive_advantage) |>
  summarise(
    snapshots = n(),
    ct_round_win_pct = 100 * mean(ct_win, na.rm = TRUE),
    .groups = "drop"
  )
show_table(advantage_by_outcome)
```

alive_advantage	snapshots	ct_round_win_pct
CT advantage	28565	83.42
Equal alive players	60682	50.68
T advantage	26954	10.40

```
p_alive_advantage <- ggplot(
  advantage_by_outcome, aes(x = alive_advantage, y = ct_round_win_pct)
) +
  geom_col() +
  geom_text(aes(label = sprintf("%.1f%%", ct_round_win_pct)), vjust = -0.3) +
  scale_y_continuous(limits = c(0, 100),
```

```
                      expand = expansion(mult = c(0, 0.08))) +
  labs(title = "CT round-win percentage by alive-player advantage",
       subtitle = "Snapshot-level descriptive comparison",
       x = "Player-count state", y = "CT round wins (%)") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 15, hjust = 1))

p_alive_advantage
```




```
save_pdf(p_alive_advantage, "22_alive_advantage_by_outcome.pdf")
```

2.3 Enriched round-state data

Check row counts and round keys

```
enrichment_comparison <- tibble(
  dataset = c("round_states", "round_states_enriched"),
  rows = c(nrow(round_states), nrow(round_states_enriched)),
  columns = c(ncol(round_states), ncol(round_states_enriched)),
  unique_rounds = c(
```

```
    nrow(distinct(round_states, match_id, round_num)),
    nrow(distinct(round_states_enriched, match_id, round_num))
  )
)
show_table(enrichment_comparison)
```

dataset	rows	columns	unique_rounds
round_states	116201	45	1257
round_states_enriched	116201	59	1257

```
snapshot_key_check <- round_states_enriched |>
  count(match_id, round_num, tick, name = "rows_per_snapshot") |>
  filter(rows_per_snapshot > 1)

tibble(duplicate_match_round_tick_keys = nrow(snapshot_key_check)) |>
  show_table()
```

duplicate_match_round_tick_keys
0

```
if (nrow(snapshot_key_check) > 0) show_table(snapshot_key_check)

unmatched_rounds <- round_states_enriched |>
  distinct(match_id, round_num) |>
  anti_join(rounds |> distinct(match_id, round_num),
            by = c("match_id", "round_num"))

tibble(enriched_rounds_not_in_rounds = nrow(unmatched_rounds)) |>
  show_table()
```

enriched_rounds_not_in_rounds
0

Missingness in enriched data

```r
enriched_missingness <- tibble(
  column = names(round_states_enriched),
  missing_values = map_int(round_states_enriched, ~ sum(is.na(.x))),
  missing_pct = 100 * missing_values / nrow(round_states_enriched)
) |>
  arrange(desc(missing_values))
show_table(enriched_missingness)
```

column	missing_values	missing_pct
seconds_since_plant	97855	84.21
plant_site	97855	84.21
plant_tick	47215	40.63
ct_won_previous_round	4116	3.54
reason_name	1263	1.09
ct_buy_type	1198	1.03
t_buy_type	1198	1.03
first_death_side	217	0.19
match_id	0	0.00
map	0	0.00
part	0	0.00
round_num	0	0.00
tick	0	0.00
seconds	0	0.00
time_remaining_s	0	0.00
bomb_planted	0	0.00
bomb_carried	0	0.00
ct_team	0	0.00
t_team	0	0.00
ct_score_before	0	0.00
t_score_before	0	0.00
ct_alive	0	0.00
ct_hp	0	0.00
ct_armor	0	0.00
ct_helmets	0	0.00
ct_defusers	0	0.00
ct_equipment	0	0.00
ct_money	0	0.00
ct_rifles	0	0.00
ct_awps	0	0.00
ct_smgs	0	0.00

column	missing_values	missing_pct
ct_smokes	0	0.00
ct_flashes	0	0.00
ct_he	0	0.00
ct_molotovs	0	0.00
t_alive	0	0.00
t_hp	0	0.00
t_armor	0	0.00
t_helmets	0	0.00
t_defusers	0	0.00
t_equipment	0	0.00
t_money	0	0.00
t_rifles	0	0.00
t_awps	0	0.00
t_smgs	0	0.00
t_smokes	0	0.00
t_flashes	0	0.00
t_he	0	0.00
t_molotovs	0	0.00
winner_side	0	0.00
ct_win	0	0.00
is_overtime	0	0.00
is_pistol_round	0	0.00
ct_equipment_freeze_end	0	0.00
t_equipment_freeze_end	0	0.00
freeze_end_tick	0	0.00
end_tick	0	0.00
round_duration_s	0	0.00
winner_team	0	0.00

Check round-level fields for consistency

These fields should generally be constant across all snapshots belonging to the same round.

```
round_level_fields <- intersect(
  c("winner_side", "winner_team", "reason_name", "is_overtime",
    "is_pistol_round", "bomb_planted", "ct_buy_type", "t_buy_type",
    "first_death_side", "round_duration_s"),
  names(round_states_enriched)
)
```

```r
round_constant_checks <- map_dfr(round_level_fields, function(field) {
  round_states_enriched |>
    group_by(match_id, round_num) |>
    summarise(
      distinct_values = n_distinct(.data[[field]], na.rm = FALSE),
      .groups = "drop"
    ) |>
    summarise(
      field = field,
      rounds_with_multiple_values = sum(distinct_values > 1),
      .groups = "drop"
    )
})
show_table(round_constant_checks)
```

field	rounds_with_multiple_values
winner_side	0
winner_team	0
reason_name	0
is_overtime	0
is_pistol_round	0
bomb_planted	626
ct_buy_type	0
t_buy_type	0
first_death_side	0
round_duration_s	0

3. Initial descriptive characteristics

Overall summary

```r
overall_summary <- rounds |>
  summarise(
    matches = n_distinct(match_id),
    rounds = n(),
    maps = n_distinct(map),
    teams_ct = n_distinct(ct_team),
    teams_t = n_distinct(t_team),
    planted_rounds = sum(bomb_planted %in% TRUE, na.rm = TRUE),
```

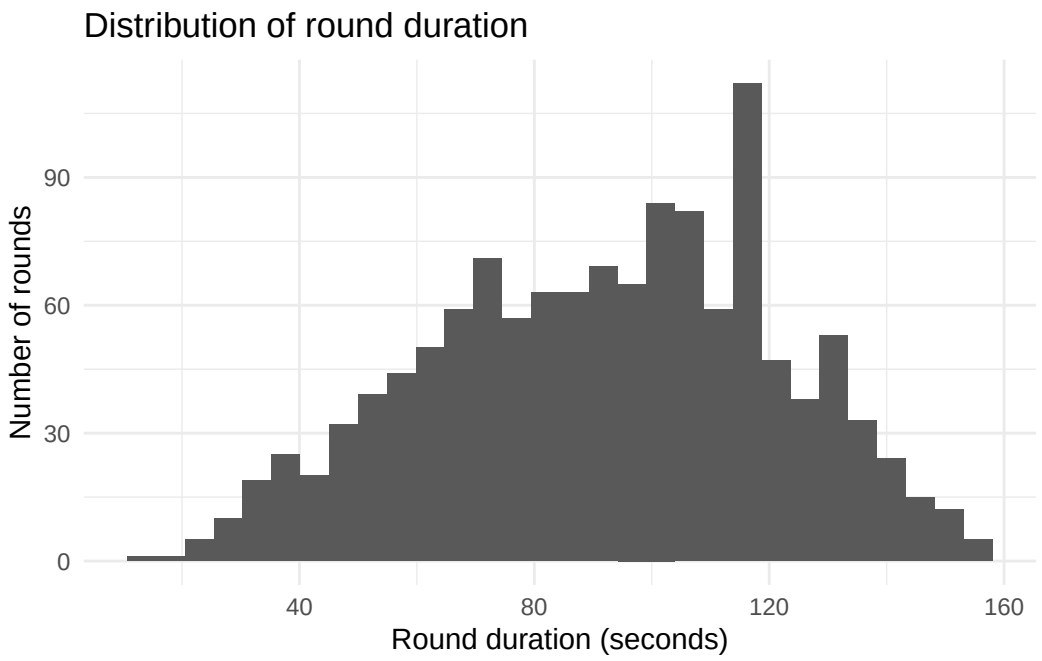
```
    planted_round_pct = 100 * mean(bomb_planted %in% TRUE, na.rm = TRUE),
    pistol_rounds = sum(is_pistol_round %in% TRUE, na.rm = TRUE),
    overtime_rounds = sum(is_overtime %in% TRUE, na.rm = TRUE),
    mean_round_duration_s = mean(round_duration_s, na.rm = TRUE),
    median_round_duration_s = median(round_duration_s, na.rm = TRUE)
  )
show_table(overall_summary)
```

matches	rounds	maps	teams_ct	teams_t	planted_rounds	planted_round_pct	pistol_rounds	ove
60	1257	2	12	12	661	52.59	120	

Round duration distribution

```
p_duration <- ggplot(
  rounds |> filter(!is.na(round_duration_s)),
  aes(x = round_duration_s)
) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of round duration",
       x = "Round duration (seconds)", y = "Number of rounds") +
  theme_minimal()

p_duration
```




```
save_pdf(p_duration, "03_round_duration_distribution.pdf")
```

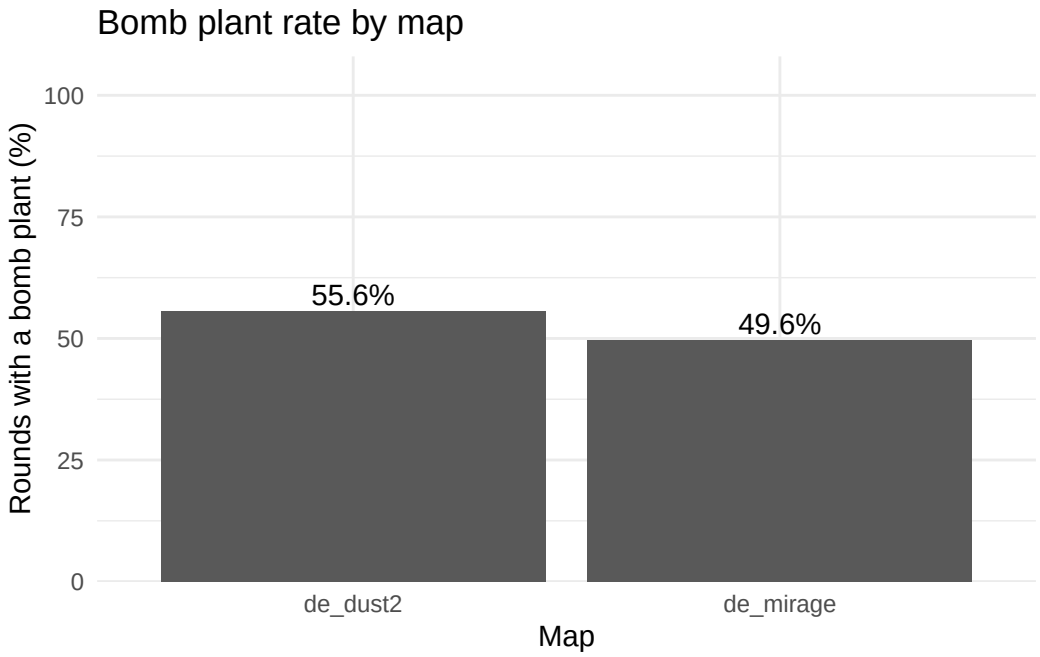
Bomb plant rate by map

```
plant_summary <- rounds |>
  group_by(map) |>
  summarise(
    total_rounds = n(),
    planted_rounds = sum(bomb_planted %in% TRUE, na.rm = TRUE),
    plant_rate_pct = 100 * mean(bomb_planted %in% TRUE, na.rm = TRUE),
    .groups = "drop"
  )
show_table(plant_summary)
```

map	total_rounds	planted_rounds	plant_rate_pct
de_dust2	628	349	55.57
de_mirage	629	312	49.60

```
p_plant_rate <- ggplot(plant_summary, aes(x = map, y = plant_rate_pct)) +
  geom_col() +
  geom_text(aes(label = sprintf("%.1f%%", plant_rate_pct)), vjust = -0.3) +
  scale_y_continuous(limits = c(0, 100),
                     expand = expansion(mult = c(0, 0.08))) +
  labs(title = "Bomb plant rate by map", x = "Map",
       y = "Rounds with a bomb plant (%)") +
  theme_minimal()

p_plant_rate
```




```
save_pdf(p_plant_rate, "03_bomb_plant_rate.pdf")
```

Buy-type distribution by map and side

```
buy_type_summary <- bind_rows(
  rounds |> count(map, buy_type = ct_buy_type, name = "rounds") |>
    mutate(side = "CT"),
  rounds |> count(map, buy_type = t_buy_type, name = "rounds") |>
    mutate(side = "T")
```

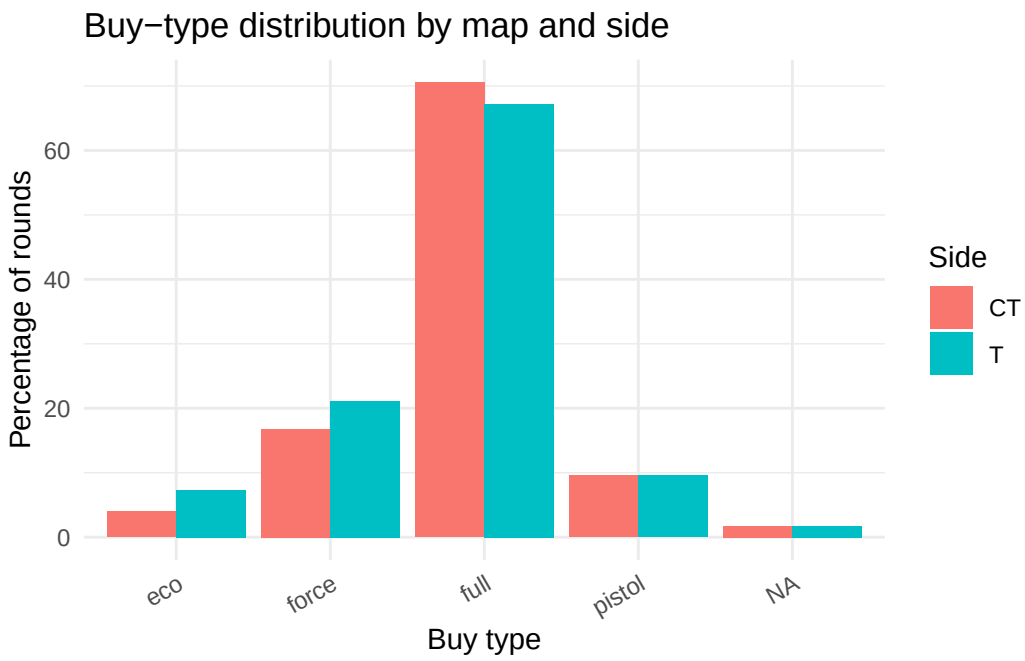
```
) |>
  group_by(map, side) |>
  mutate(percentage = 100 * rounds / sum(rounds)) |>
  ungroup()

show_table(buy_type_summary)
```

map	buy_type	rounds	side	percentage
de_dust2	eco	20	CT	3.18
de_dust2	force	105	CT	16.72
de_dust2	full	443	CT	70.54
de_dust2	pistol	60	CT	9.55
de_mirage	eco	25	CT	3.97
de_mirage	force	94	CT	14.94
de_mirage	full	439	CT	69.79
de_mirage	pistol	60	CT	9.54
de_mirage	NA	11	CT	1.75
de_dust2	eco	46	T	7.32
de_dust2	force	100	T	15.92
de_dust2	full	422	T	67.20
de_dust2	pistol	60	T	9.55
de_mirage	eco	44	T	7.00
de_mirage	force	133	T	21.14
de_mirage	full	381	T	60.57
de_mirage	pistol	60	T	9.54
de_mirage	NA	11	T	1.75

```
p_buy_types <- ggplot(
  buy_type_summary, aes(x = buy_type, y = percentage, fill = side)
) +
  geom_col(position = "dodge") +
  labs(title = "Buy-type distribution by map and side",
       x = "Buy type", y = "Percentage of rounds", fill = "Side") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 30, hjust = 1))

p_buy_types
```




```
save_pdf(p_buy_types, "03_buy_type_distribution.pdf")
```

4. Initial analysis ideas

These comparisons are descriptive, not causal. For round-level comparisons, use one observation per round. The state-based analyses below select the latest snapshot at or before 30 seconds.

4.1 Bomb plant and round outcome

```
plant_outcomes <- rounds |>
  mutate(
    plant_status = case_when(
      bomb_planted %in% TRUE ~ "Planted",
      bomb_planted %in% FALSE ~ "Not planted",
      TRUE ~ "Unknown"
    )
  ) |>
  group_by(map, plant_status) |>
  summarise(
    rounds = n(),
```

```
    ct_win_pct = 100 * mean(ct_win, na.rm = TRUE),
    .groups = "drop"
  )
show_table(plant_outcomes)
```

map	plant_status	rounds	ct_win_pct
de_dust2	Not planted	279	83.51
de_dust2	Planted	349	22.64
de_mirage	Not planted	317	86.12
de_mirage	Planted	312	25.00

```
plant_site_outcomes <- rounds |>
  filter(bomb_planted %in% TRUE, !is.na(plant_site)) |>
  group_by(map, plant_site) |>
  summarise(
    planted_rounds = n(),
    ct_win_pct = 100 * mean(ct_win, na.rm = TRUE),
    .groups = "drop"
  )
show_table(plant_site_outcomes)
```

map	plant_site	planted_rounds	ct_win_pct
de_dust2	A	199	26.63
de_dust2	B	150	17.33
de_mirage	A	187	31.55
de_mirage	B	125	15.20

4.2 Alive-player advantage at 30 seconds

```
state_at_30s <- round_states_enriched |>
  filter(seconds <= 30) |>
  group_by(match_id, round_num) |>
  slice_max(order_by = seconds, n = 1, with_ties = FALSE) |>
  ungroup() |>
  mutate(
    alive_difference = ct_alive - t_alive,
    alive_advantage = case_when(
```

```
      alive_difference > 0 ~ "CT has more alive",
      alive_difference < 0 ~ "T has more alive",
      alive_difference == 0 ~ "Equal alive players",
      TRUE ~ NA_character_
    )
  )

alive_outcome_30s <- state_at_30s |>
  filter(!is.na(ct_win), !is.na(alive_advantage)) |>
  group_by(alive_advantage) |>
  summarise(
    rounds = n(),
    ct_win_pct = 100 * mean(ct_win, na.rm = TRUE),
    .groups = "drop"
  )
show_table(alive_outcome_30s)
```

alive_advantage	rounds	ct_win_pct
CT has more alive	300	81.00
Equal alive players	749	50.73
T has more alive	208	19.23

4.3 Equipment and economy advantage at 30 seconds

```
economy_summary <- round_states_enriched |>
  mutate(
    equipment_difference = ct_equipment - t_equipment,
    money_difference = ct_money - t_money
  ) |>
  group_by(match_id, round_num) |>
  filter(seconds <= 30) |>
  slice_max(order_by = seconds, n = 1, with_ties = FALSE) |>
  ungroup() |>
  filter(
    !is.na(ct_win),
    !is.na(equipment_difference),
    !is.na(money_difference)
  ) |>
  mutate(
```

```
    equipment_advantage = case_when(
      equipment_difference > 0 ~ "CT higher equipment",
      equipment_difference < 0 ~ "T higher equipment",
      equipment_difference == 0 ~ "Equal equipment"
    )
  ) |>
  group_by(equipment_advantage) |>
  summarise(
    rounds = n(),
    ct_win_pct = 100 * mean(ct_win, na.rm = TRUE),
    mean_equipment_difference = mean(equipment_difference, na.rm = TRUE),
    mean_money_difference = mean(money_difference, na.rm = TRUE),
    .groups = "drop"
  )
show_table(economy_summary)
```

equipment_advantage	rounds	ct_win_pct	mean_equipment_difference	mean_money_difference
CT higher equipment	793	68.73	9770.11	5265.07
Equal equipment	6	50.00	0.00	-616.67
T higher equipment	458	25.11	-8162.12	-5150.87

4.4 Health and armor advantage at 30 seconds

```
health_armor_summary <- round_states_enriched |>
  mutate(
    hp_difference = ct_hp - t_hp,
    armor_difference = ct_armor - t_armor
  ) |>
  group_by(match_id, round_num) |>
  filter(seconds <= 30) |>
  slice_max(order_by = seconds, n = 1, with_ties = FALSE) |>
  ungroup() |>
  filter(!is.na(ct_win)) |>
  mutate(
    hp_advantage = case_when(
      hp_difference > 0 ~ "CT higher HP",
      hp_difference < 0 ~ "T higher HP",
      hp_difference == 0 ~ "Equal HP",
      TRUE ~ NA_character_
```

```
    )
  ) |>
  group_by(hp_advantage) |>
  summarise(
    rounds = n(),
    ct_win_pct = 100 * mean(ct_win, na.rm = TRUE),
    mean_hp_difference = mean(hp_difference, na.rm = TRUE),
    mean_armor_difference = mean(armor_difference, na.rm = TRUE),
    .groups = "drop"
  )
show_table(health_armor_summary)
```

hp_advantage	rounds	ct_win_pct	mean_hp_difference	mean_armor_difference
CT higher HP	551	68.42	95.50	120.13
Equal HP	342	51.75	0.00	27.62
T higher HP	364	29.95	-76.66	-84.38

Notes and interpretation

- `rounds.parquet` contains round-level records; `round_states.parquet` contains multiple time snapshots per round.
- `round_states_enriched.parquet` adds round-level information to each snapshot. The row count should normally remain the same as the original snapshot data.
- Missing `seconds_since_plant` and `plant_site` values may be expected in rounds or snapshots where the bomb has not been planted.
- The snapshot-level alive-advantage summary in section 2.2 weights rounds with more snapshots more heavily. Use the one-snapshot-per-round analysis in section 4.2 for a more comparable round-level description.
- The 30-second analyses use the latest available snapshot at or before 30 seconds; it may not be exactly 30 seconds.
- Figures are saved as PDF files in `../reports/figures`.